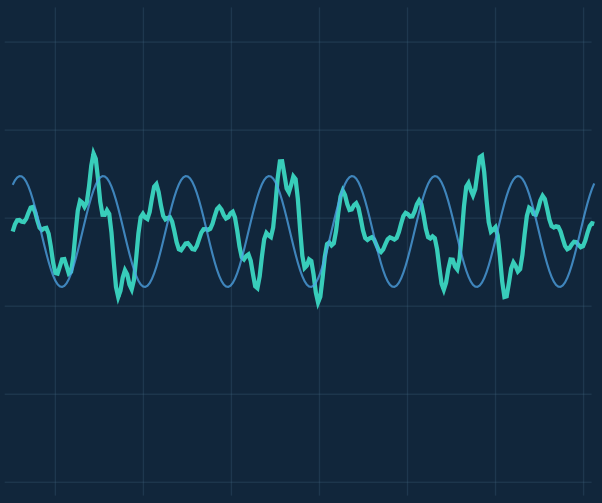
CLOUD COMPUTING / DEEP LEARNING

VibeGuard Cloud

Load-Shift-Aware Motor Bearing Fault Detection using Deep Learning and AWS

1D CNN | AWS EC2 | AWS S3



THE PROJECT IN ONE PARAGRAPH

VibeGuard Cloud is a **cloud-hosted research prototype** that analyzes electric-motor vibration recordings to identify likely bearing faults. A small deep learning model examines signal patterns, returns one of four operating conditions, and flags uncertain results for human review. The application is hosted on AWS and explored through a simple web dashboard.

WHY ARE WE BUILDING IT?

Industrial motors are critical to factories and machinery. Unexpected bearing problems can lead to costly interruptions, while manual inspection of vibration data is time-consuming.

Goal: demonstrate how cloud infrastructure and deep learning could support faster, more consistent preliminary fault analysis.

WHAT MAKES IT DIFFERENT?

Rather than measuring accuracy only on familiar operating conditions, the project evaluates whether the AI works at a **motor load it has not seen during training**.

It also compares a CNN with a traditional model and highlights cases needing human review.

FOUR CONDITIONS THE MODEL ANALYZES

01	02	03	04
NORMAL	INNER RACE	BALL FAULT	OUTER RACE

RESEARCH SCOPE

The prototype uses recorded laboratory vibration data, not an installed live sensor. It detects predefined **bearing conditions**; it does not calculate remaining useful life or replace a certified maintenance decision.

SYSTEM FUNCTIONING

How VibeGuard Cloud works

From raw vibration recordings to visualized model results in a cloud-based dashboard.

01

Collect vibration data

Use the public Case Western Reserve University (CWRU) bearing dataset containing normal and experimentally induced bearing-fault recordings.

02

Prepare the signal

Extract the drive-end vibration channel, split it into short 2,048-sample windows and normalize each window before analysis.

03

Train the deep learning model

A 1D Convolutional Neural Network (CNN) learns patterns linked to the four bearing conditions. A Random Forest provides a comparison baseline.

04

Test on a different motor load

Train with 0-1 HP recordings, validate with 2 HP, and test with previously unseen 3 HP recordings to investigate generalization.

05

Display predictions and review flags

Upload a recording in Streamlit. View predicted condition, class scores, vibration waveform, segment agreement and a manual-review flag.

06

Operate in the cloud

Use AWS EC2 and EBS for model execution and the application; store datasets, model files and evaluation outputs privately in Amazon S3.

CLOUD ARCHITECTURE AT A GLANCE

CWRU data > **Signal processing + 1D CNN** > **Streamlit dashboard**

Runs on AWS EC2 + EBS | Private model and dataset storage: Amazon S3

Expected final output: A functioning AWS-hosted demo, trained model, CNN-vs-baseline comparison, held-out-load evaluation, saved artifacts, and a project report. **Actual accuracy will be reported only after training and testing.**

Dataset reference: Case Western Reserve University Bearing Data Center - engineering.case.edu/bearingdatacenter